MILITARY SPECIFICATION

HEATER, IMMERSION, M67, LIQUID FUEL FIRED, FOR CORRUGATED CANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a liquid fuel burning immersion-type heater designed primarily to heat water in a 32-gallon corrugated can.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning Solvent.
GG-P-455	-	Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
QQ-B-626	-	Brass, Leaded and Nonleaded: Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-654	-	Brazing Alloys, Silver.
QQ-C-390	-	Copper Alloy Castings (Including Cast Bar).
QQ-I-652	-	Iron Castings, Gray.
QQ-S-635	-	Steel Plate, Carbon.
QQ-S-698	-	Steel Sheet and Strip, Low-Carbon.
QQ-S-777	-	Steel, Carbon, Strip, Cold-Rolled, Untempered Spring Quality.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-S-781 - Strapping, Steel, and Seals.
QQ-W-321 - Wire, Copper Alloy.
QQ-W-423 - Wire, Steel, Corrosion-Resisting.
QQ-Z-325 - Zinc Coating, Electrodeposited Requirements For.
RR-C-82 - Cans, Ash and Garbage, Corrugated, Taper-Side, Zinc-Coated: And Covers.
RR-C-271 - Chains and Attachments, Welded and Weldless.
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coating.
TT-E-529 - Enamel, Alkyd, Semigloss.
TT-P-636 - Primer Coating, Alkyd, Wood and Ferrous Metal.
WW-P-406 - Pipe, Steel (Seamless and Welded) (For Ordinary Use).
WW-P-521 - Pipe Fittings, Flange Fittings, and Flanges Steel and Malleable Iron (Threaded and Butt-Welded), 150 Pound.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-551 - Pipes and Elbows, Air Conditioning - Heating.
MIL-R-3065 - Rubber, Fabricated Products.
MIL-L-3150 - Lubricating Oil, Preservative, Medium.
MIL-C-10382 - Corrosion Preventive, Petrolatum, Spraying Application: for Food Handling Machinery and Equipment.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative and Break-In.

STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging, and Packing Materials: Test Procedures.
FED-STD-595 - Colors.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-417 - Classification System and Tests for Solid Elastomeric Materials (Symbols and Tests).

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-1874 - Heater, Immersion, M67, Liquid Fuel Fired, Cans, Corrugated; Containers for Level "A" Shipment.
5-13-2679 - Assembly Complete.
5-13-2680 - Heater Body Assembly.
5-13-2681 - Heater Body Details.
5-13-2682 - Miscellaneous Parts.
5-13-2683 - Burner Assembly.
5-13-2684 - Igniter Assembly and Name Plate.
5-13-2685 - Fuel Tank Assembly.
5-13-2686 - Valve and Filler Plug Assembly.
5-13-2687 - Hanger Details.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw - Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

AMERICAN WELDING SOCIETY CODE

Standard Qualification Procedure of the American Welding Society (AWS)

(Application for copies of the American Welding Society Code should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEER'S CODE

Welding Qualification of the American Society of Mechanical Engineers (ASME)

(Application for copies of the ASME Codes should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10018.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

AMERICAN SOCIEITY FOR TESTING AND MATERIALS (ASTM)

	A 167	-	Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
*	A 511	-	Seamless Stainless Steel Mechanical Tubing
	A 513	-	Electric-Resistance Welded Carbon and Alloy Steel Tubing
	A 519	-	Seamless Carbon and Alloy Steel Mechanical Tubing
*	A 575	-	Merchant Quality Hot-Rolled Carbon Steel Bars
*	A 684	-	Steel, Carbon, Strip, Cold Rolled Soft, Untempered Spring Quality

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

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3.2 Materials and components. Materials and components specified herein and on the drawings shall conform to the following applicable requirements.

3.2.1 Steel, carbon.

3.2.1.1 Steel, carbon sheet and strip. Carbon steel sheet and strip shall conform to QQ-S-698.

* 3.2.1.2 Bar stock. Cold finished carbon steel stock shall conform to QQ-S-635. Hot rolled steel bar stock shall conform to ASTM A 575.

3.2.1.3 Seamless steel tubing. Hot finished or cold worked finished seamless steel tubing shall conform to ASTM A 519.

3.2.1.4 Welded steel tubing. As-welded hot or cold rolled steel tubing shall conform to ASTM A 513.

* 3.2.1.5 Spring steel. Spring carbon steel strip shall conform to soft type, annealed, of ASTM A 684.

3.2.2 Corrosion-resisting steel.

3.2.2.1 Stainless steel sheet and strip. Stainless steel sheet and strip shall conform to type 304 and type 305 of ASTM A 167.

3.2.2.2 Corrosion-resisting steel wire. Corrosion-resisting steel wire shall conform to condition B, of QQ-W-423.

3.2.3 Stainless steel tubing. Stainless steel tubing shall conform to grade 304 of ASTM A 511.

3.2.4 Cast iron. Cast iron shall conform to the mechanical properties of class 25 of QQ-I-652.

3.2.5 Brass.

* 3.2.5.1 Brass rod. Brass rod shall conform to 1/2 hard copper alloy No. 360 of QQ-B-626.

3.2.5.2 Brass wire. Brass wire shall conform to 1/2 hard of QQ-W-321.

* 3.2.5.3 Brass casting. Brass casting shall conform to copper alloy No. 851-860 of QQ-C-390.

3.2.6 Brazing alloy. Brazing alloy shall conform to QQ-B-654.

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3.2.7 Gaskets.

3.2.7.1 Filler plug. The filler plug gasket shall be lead sheet conforming to grade B, of QQ-L-201.

3.2.7.2 Vent plug. The vent plug gasket shall be synthetic rubber conforming to SB615 AE5 of MIL-R-3065 and MIL-STD-417.

3.2.8 Nipple. The fuel supply line nipple shall conform to weight A, class 1 of WW-P-406.

3.2.9 Elbow. The fuel supply line elbow shall conform to type I of WW-P-521.

3.2.10 Safety chain. The brass safety chain shall conform to type II, class 6, 27 links per foot of RR-C-271.

3.2.11 Asbestos.

3.2.11.1 Valve packing. The asbestos used in the fabrication of the valve packing shall be asbestos rope made from long-fiber carded asbestos, spun into yarn and laid up into the general form of a loosely twisted cord impregnated with graphite. If a core is employed, it shall be of the same material and construction.

3.2.11.2 Igniters. The asbestos wick for the igniter shall be soft, highly absorbent, long-fiber asbestos in the form of a braided rope.

3.2.12 Cotter pin. The cotter pin for the igniter assembly hinge shall be cadmium plated steel.

* 3.2.13 Stove pipe. Stove pipe shall have a blued finish and shall conform to style 4, size 4-inch diameter of MIL-P-551. Four lengths of stove pipe shall be furnished with each immersion heater.

3.2.14 Primer. Primer shall conform to TT-P-636.

3.2.15 Enamel. Enamel shall conform to class A or B of TT-E-529. The color shall approximate color number 24087, olive drab of FED-STD-595.

3.2.16 Photographic plate. The photographic nameplate on the heater body and the instruction plate on the hood shall conform to type I, class 1 of GG-P-455 (see 6.4). The photographic nameplate and the caution plate on the fuel tank may be either plain or backed with a pressure sensitive adhesive (see 6.5). The caution plate color shall be predominately yellow.

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3.3 Design and construction. The immersion heater components and assembly shall be constructed in accordance with Drawings 5-13-2679 through 5-13-2687 and requirements specified herein. Spot welds shall be adherent and well made so that the contact faces of the joint cannot be pulled apart without tearing the metal when tested as specified in 4.6.1. The nameplate on the fuel tank as shown on Drawing 5-13-2684 shall be applied smoothly and completely adhered to the tank.

3.3.1 Threads. Threads shall conform to National Bureau of Standard Handbook H28.

3.3.2 Fuel tank. The fuel tank assembly shall be welded as specified except for the fuel line and vent valve bushings. The fuel line and vent valve bushings shall be brazed with brazing alloy specified in 3.2.6. The inside of the fuel tank assembly shall be cleaned of all foreign matter such as welding splatter, grease and millings. The tank assembly prior to application of finish shall not leak under 4 $\pm$ 1/2 psig (pounds per square inch, gage) air pressure when tested in accordance with 4.6.2.

3.3.3 Heater body. The heater body prior to application of finish shall not leak under 4 $\pm$ 1/2 psig air pressure when tested in accordance with 4.6.3.

3.3.4 Igniter cup. Igniter cup shall not leak when tested in accordance with 4.6.4.

3.3.5 Filler plug. The filler plug gasket as specified in 3.2.7.1 shall be assembled to the filler plug with a press fit.

* 3.3.6 Valve assembly. The valve shall meter and stop the flow of fuel when tested in accordance with 4.6.6. The valve stem shall not bind throughout its entire engagement. The valve shall seat and stop the flow of air when closed with a torque of less than 4.0 inch pounds.

3.4 Welder qualification. The welders performing the welding on sub-assemblies, assemblies and on the end item shall be certified welders and have passed the qualification test as prescribed by either the "Standard Qualification Procedures of the American Welding Society" or "Welding Qualification of the American Society of Mechanical Engineers" code. The contractor shall furnish the Government with a list of names of his welders who are certified and have passed the test of either referenced codes and he shall certify that the welding of this equipment was performed by the welders listed.

3.5 Finish.

3.5.1 Zinc plating. Prior to assembly, zinc plating shall conform to type II, class 2 of QQ-Z-325.

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3.5.2 Priming and painting. Prior to priming, surfaces of the heater assembly shall be cleaned in accordance with TT-C-490. Cleaned surfaces shall be coated with primer specified in 3.2.14, thoroughly dried and coated with enamel specified in 3.2.15. When class A enamel is used, the finish may be forced dried. When class B enamel is used, the enamel shall be baked. Touch-up of bare spots and run-over inside of flue and burner tubes is acceptable. Surfaces of the heater assembly shall be painted as follows:

* (a) The heater body assembly down to a horizontal line 6 inches below the top of the burner tube that includes:

(1) The flue tube (prior to the assembly of the igniter assembly).
(2) The burner tube (prior to the assembly of the hood).
(3) The hanger support.
(4) The hanger assembly except that the hanger arm shall be completely painted exclusive of the screw threads.

(b) All surfaces of the backing plate assembly.
(c) Igniter assembly - All surfaces except wick and retainer spring.
(d) Fuel tank - Except for the drip valve, all outside surfaces of the fuel tank including the filler plug and vent plug in the closed position.
(e) Hood - Prior to assembly of instruction plate, all surfaces of the hood including exposed surfaces of the hinges.

3.6 Instruction and nameplates. Instruction and nameplates shall consist of photographic plate specified in 3.2.16 (see 6.4). Dimensions and lettering shall conform to Drawing 5-13-2684.

3.7 Workmanship.

3.7.1 Fabrication. The immersion heater assembly shall be free of burrs, slivers, dents, and fractures. The fuel tank when mounted on the heater, shall be in horizontal and vertical alignment with flue and burner tubes. The cover of the igniter assembly shall lay flush against the flue tube to completely cover the burner cup entrance hole.

3.7.2 Plating and painting. Plating and painting shall be smooth, adherent, free of blisters, cracks, runs, tackiness, imbedded foreign material and areas of no finish.

3.7.3 Welding and brazing. The surface of the parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free from cracks, burnholes, undercuts, splatter or incomplete fusion. All scale and flux shall be removed from the welded area.

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3.7.4 Spot welding. Spot welds, when applicable, shall be free of splatter, and burn through and not otherwise poorly made.

3.7.5 Rivets. Rivets shall be tightly staked and shall not be broken.

* 3.8 Performance. The immersion heater, when operated in accordance with operating instructions on the nameplate (see Drawing 5-13-2684) and at an ambient temperature from 60° to 80°F shall be capable of burning 86 to 94 antiknock index grade automotive gasoline (leaded or unleaded) to heat 24 gallons of water through a 100°F temperature rise within 35 minutes when tested as specified in 4.6.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required, the immersion heater shall be examined as specified in 4.4.3.1 and 4.4.3.2 and tested as specified in 4.6.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.4.2 In-process inspection.

4.4.2.1 Visual examination of heater surface preparation. Prior to enameling, examination shall be made of cleaning and priming of heater surfaces to establish conformance with requirements specified in 3.5.2. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

4.4.2.2 Spot weld test. Prior to spot welding, the spot welding process shall be tested as specified in 4.6.1. Any failure to pass the test shall be cause for correction of the process.

4.4.2.3 Leak test. Prior to assembly, each fuel tank, heater body and igniter cup shall be tested for leaks as specified in 4.6.2 through 4.6.4, respectively. Failure to pass any test shall be cause for rejection of the affected item.

4.4.3 End item inspection. The inspection lot shall be all immersion heaters offered for inspection at one time. The sample unit shall be a complete immersion heater.

4.4.3.1 Visual examination. The immersion heaters shall be examined for the defects in table I. The inspection level shall be II with an AQL (acceptable quality level) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Missing, not as specified; finish coat not smooth, free of blisters, cracks, runs, tackiness, areas of no film and imbedded foreign matter, or otherwise impaired		X
Materials, construction and workmanship (applicable to all components, subassemblies and assemblies)	Component material not as specified	X	
	Any part missing or not specified type	X	
	Fractured, split, punctured torn, dented, creased, deteriorated, bowed, sprung, malformed, miscast	X	
	Component misplaced, leaking or not in proper alignment	X	
	Evidence of sharp burrs, slivers, or edges	X	
	Any functioning component that requires abnormal force to operate		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Casting	Not connected or joined as specified or poorly accomplished	X	
Welding and brazing	Not free from rust, flux, scale, paint, grease or other foreign matter		X
	Not smooth and free from cracks, burnholes, undercuts, splatter or incomplete fusion	X	
Spot welds	Not free of splatter and burn through, or otherwise poorly made	X	
Bolts, nuts, screws, studs, and other threaded fastener components	Missing, not type specified, broken, stripped or fractured	X	
	Loose		X
Rivets	Missing, not tightly staked or broken	X	
Pipe, tubing, fittings and valves	Improperly connected, installed, or stripped	X	
Assembly	Any component not properly assembled or secured	X	
	Unit perceptibly out of square or alignment	X	
Identification and instruction marking	Missing, incomplete, illegible, not specified type or size, not in proper location, or not affixed in prescribed manner	X	
	Fuel tank nameplate not smooth or in complete adherence with tank		X

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4.4.3.2 Dimensional examination. The immersion heater shall be examined for compliance with dimensions specified. Any dimension not within the specified tolerances shall be a defect. The inspection level shall be S-1 with an AQL of 6.5 defects expressed in terms of defects per hundred units.

* 4.4.3.3 End item testing. When a first article is not required, the initial production unit shall be tested as specified in 4.6.5. Failure to pass the test shall be cause for rejection of the initial production unit.

4.4.4 Examination of packaging. An examination shall be made to determine that preservation-packaging, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. The lot shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of packaging

Examine	Defect
Markings, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Preservative improperly applied or missing
Materials and components	Component missing, damaged or otherwise defective closure or container flaps, loose strapping, or inadequate stapling; bulging or distortion of container; blocking and cushioning inadequate, improper or missing

4.5 Certificate examination. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

4.6 Tests.

4.6.1 Spot weld test. Test coupons of the mating metals shall be joined together with spot welds in the same manner as intended for the parent components. The joint contact faces shall be pulled apart to determine compliance with the spot weld requirements of 3.3. Any nonconformance shall constitute failure of this test.

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4.6.2 Fuel tank test. The fuel tank shall be pressurized with air at 4 ± 1/2 psig and tested with a bubble supporting film in accordance with Method 5009 of FED-STD-101, to determine compliance with the leak requirements of 3.3.2. Any nonconformance shall constitute failure of this test.

4.6.3 Heater body test. The immersion heater body shall be pressurized with air at 4 ± 1/2 psig and tested with a bubble supporting film in accordance with Method 5009 of FED-STD-101 to determine compliance with the leak requirements of 3.3.3. Any nonconformance shall constitute failure of this test. At the completion of this test the heater body shall be thoroughly dried.

4.6.4 Igniter cup test. The igniter cup shall be filled with a fluid and observed for leaks to determine compliance with 3.3.4. Any nonconformance shall constitute failure of this test.

* 4.6.5 Performance test. The immersion heater shall be installed with the stack plumb in a 32 gallon corrugated can conforming to RR-C-82 and containing 24 gallons of water with a thermometer located in the center of the can opening and the bottom of the bulb immersed a minimum of 3 inches below the surface of the water. The heater shall be started with the fuel specified in 3.8 and operated at the steady fuel rate obtained by constantly adjusting the fuel valve so that smoke from the stack remains barely visible throughout this test. The valve shall meter and stop the flow of fuel and not bind throughout its entire engagement as specified in 3.3.6. The time to heat the water shall be checked to determine compliance with the water heating requirements of 3.8. Any nonconformance shall constitute failure to pass this test.

* 4.6.6 Valve assembly test. The valve assembly will be pressurized at 4 ± 1/2 psig to check for compliance with 3.3.6 to insure that the valve shall seat and stop the flow of air when closed with a torque of less than 4 inch pounds as specified in 3.3.6. This test may be combined with 4.6.2.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Each heater shall have the fuel tank detached from the body of the heater. The four stovepipe lengths shall be detached from the heater and disassembled. The burner shall be removed.

5.1.1.2 Cleaning. All uncoated ferrous metal surfaces of the heater and detached components shall be cleaned with dry-cleaning solvent conforming to P-D-680 and thoroughly dried prior to application of preservative.

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5.1.1.3 Preservation. The interior of the fuel tank and interior of the heater body shall be fogged with preservative conforming to type 1 of MIL-L-21260 and all excess preservative drained off. All exterior uncoated surfaces of the heater body, including hanger and hanger screw threads, and fuel tank valve stem assembly shall be coated with preservative conforming to MIL-C-10382. The burner shall be completely wrapped with grease-proof barrier-material conforming to type II, grade A of MIL-B-121. The four cleaned lengths of the stovepipe shall be individually coated with medium preservative oil conforming to MIL-L-3150.

5.1.1.4 Packaging.

5.1.1.4.1 Heater body. The wrapped burner shall be placed within the burner flue and the hood shall be sealed in place over the burner flue with tape conforming to type IV, class 1 of PPP-T-60. The hanger screws shall be turned in, aligned, and secured to form the smallest package possible.

5.1.1.4.2 Fuel tank. The valve assembly of the fuel tank shall be wrapped with grease-proof barrier-material specified in 5.1.1.3 and secured with tape as specified in 5.1.1.4.1.

5.1.1.4.3 Stovepipe. The rivets used in the stovepipe lengths as retaining means shall be placed in a cotton cloth drawstring bag and securely tied to the outer stovepipe through one of the rivet holes. The preserved stovepipe lengths shall be nested one within the other and wrapped with grease-proof barrier-material specified in 5.1.1.3. The wrapped stovepipe lengths shall then be packaged in a snug-fitting end-opening fiberboard box conforming to style FOL, W5c of PPP-B-636. The box shall be closed, waterproofed and reinforced by means of tape in accordance with the appendix of PPP-B-636.

5.1.1.4.4 Placement. The fuel tank shall by placed on end, the filler plug up, on top of the lower right portion of the heater body and against the flue stack so that the fuel tank support shall be located in one of the corners of the shipping container. The packaged stovepipe shall be placed adjacent to the flue and the fuel tank on top of the lower left portion of the heater body. The tank and packaged stovepipe length shall then be secured in place by not less than three bands of tape minimum 1/2 inch wide conforming to type IV of PPP-T-97. The tape shall be applied in a manner as to preclude loosening, movement, or damage to the secured components.

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* 5.1.2 Commercial. Heaters shall be preserved with contact non-toxic preservatives and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each heater, complete with components, preserved and packaged as specified in 5.1, shall be packed in a shipping container conforming to PPP-C-650 and nailed wood container illustration shown on Drawing 5-13-1874, or a wirebound container conforming to PPP-B-585 and wirebound container illustration on Drawing 5-13-1874. Each heater shall be positioned within the shipping container with the flue in the same corner as the block provided on each container base. Heaters packed in wood crates shall be strapped to the base by means of a 3/4 by 0.023-inch flat steel strap conforming to class 1, type I of QQ-S-781. The container top shall be secured in place so that the block provided shall engage the exhaust flue tube opening. Each shipping container shall be closed and reinforced with flat steel strapping in accordance with the appendix of the applicable container specification.

5.2.2 Commercial packing. Heaters, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The immersion heater is intended for heating or boiling water for cleaning kitchen utensils of field kitchens and mess kits and for other general uses.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When first article is not required (see 3.1).
(c) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

* 6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Photographic plates. Photographic plates for heater body and hood by Metalphoto Corporation, 18531 South Miles Road, Cleveland, OH 44128 or equal are satisfactory (see 3.2.16).

6.5 Pressure sensitive plates. 3M Scotch Cal photosensitive plate number 8005 by the 3M Company, St. Paul, MN or equal is satisfactory for use as nameplate and caution plate with pressure sensitive adhesive, on the fuel tank. Industrial adhesives: Pliobond No. 20 by Goodyear Rubber and Tire Co., Akron, OH 44316; A851B by B. F. Goodrich General Products Company, Division of B. F. Goodrich Company, Akron, OH 43318 and Scotch-Grip by 3M Adhesives, Coatings and Sealers Division, 3M Center, St. Paul, MN 55101 or equal are satisfactory for use in adhering plain photographic nameplate on the fuel tank (see 3.2.16).

6.6 Changes from previous issue. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) were made from previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 4540-0057

Review activities:

Army - ME, MD
DSA - CS

User activities:

Army - AV, SM
Navy - MC

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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